

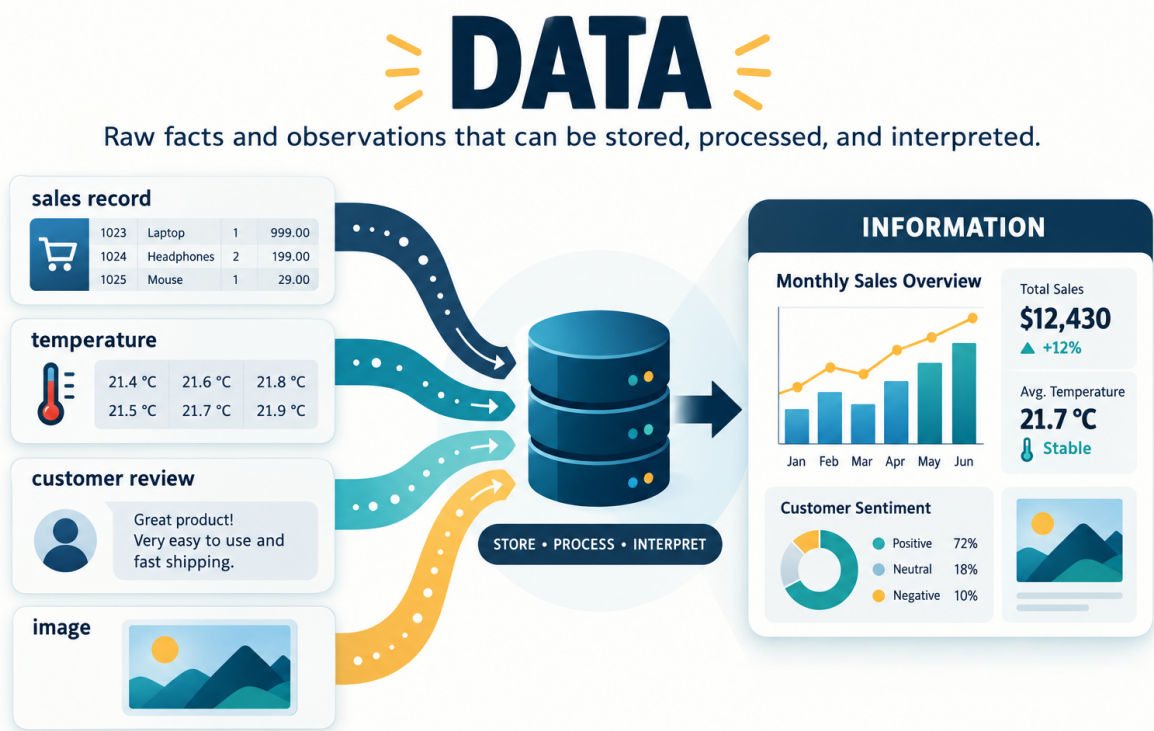
Information Systems: Data and Knowledge Management

Student reading guide

This downloadable guide follows the eight-section learning resource. Each section develops the organisational context behind the concepts, systems, infrastructure, and decisions discussed on the website.

Central question: how do organisations turn data into useful, trusted, and actionable capability?

01 From data to knowledge



Recommended video: Data vs Information vs Knowledge: The Key to Smart Thinking — Insight Lab Studios

This video explains how raw observations become meaningful information and how knowledge requires interpretation, critical thinking, and judgement. [Watch the video](#) ■

READING // ORGANISATIONAL CONTEXT

Organisations do not work with “**data**” in the abstract. They work with records created by activities: a customer places an order, a student submits an assignment, a vehicle reports its location, or a machine produces a reading. Each record describes something that happened, but it does not explain why it happened or what should happen next. The first challenge is therefore not simply collecting more records. It is deciding which observations matter, how they should be described, and what questions they may help people answer.

A useful distinction is between a recorded observation, **information**, and **knowledge**. A timestamp or percentage is data. It becomes information when it is placed in context: a particular student, date, assessment, or comparison group. Knowledge develops when someone interprets that information using experience, purpose, and judgement. This distinction matters because an organisation can have accurate data and still make poor decisions if the data is misunderstood or disconnected from the decision being considered.

Context is often supplied through definitions and relationships. “Active customer” might mean someone who bought something this month, someone with a current subscription, or someone who has logged in recently. None of these definitions is automatically correct. The appropriate meaning depends on the organisation’s purpose. If different teams use the phrase differently, their reports may appear to disagree even when they are drawing on the same underlying records.

Information systems transform data as it moves through an organisation. Data may be selected, cleaned, classified, combined, calculated, visualised, and presented to a user. A streaming service can turn clicks and viewing events into categories such as preferred genres or likely interests. A university can turn login records and assessment attempts into a report about participation. Every transformation makes the data more useful, but every transformation also introduces choices and assumptions.

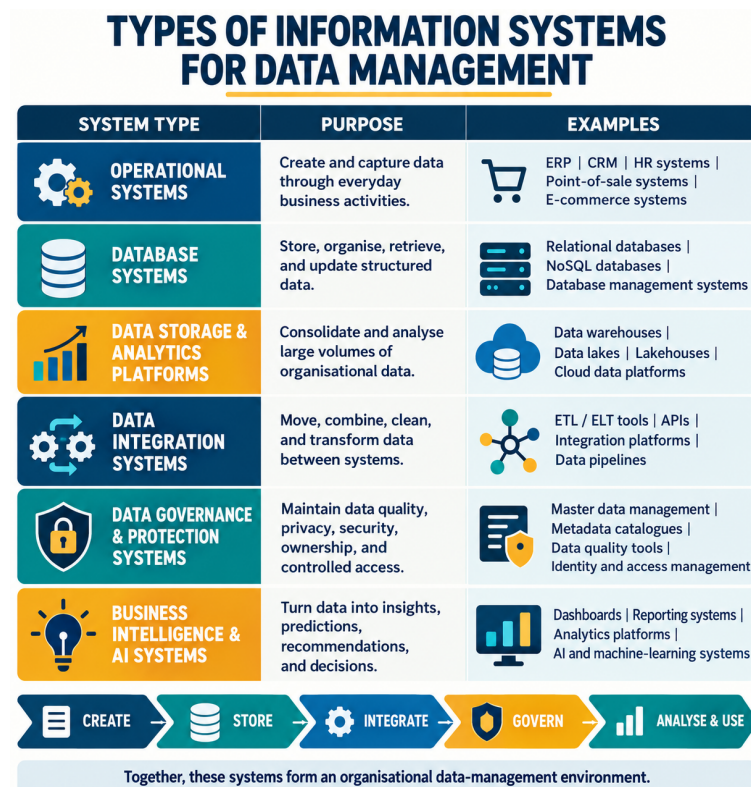
People remain central because meaning is not contained inside a database or dashboard. A manager decides whether a fall in sales is a problem, a seasonal pattern, or evidence that a measure has changed. A teacher interprets a fall in quiz attempts alongside assessment deadlines and student feedback. The system supplies evidence, but people connect that evidence to goals, responsibilities, constraints, and possible action.

The same data can support different decisions. A delivery company may use location records to estimate arrival times, plan staffing, investigate delays, or evaluate a route. Each purpose requires different context and may require a different level of accuracy or timeliness. Data that is sufficient for a broad trend may be insufficient for an individual customer promise. “Good data” is therefore always related to a use.

There are also ethical and political questions. Organisations decide what to record, whose behaviour becomes visible, which categories are used, and which outcomes count as success. A recommendation system may make a service easier to use while also shaping what users see. A risk score may support consistent decisions while reproducing the limitations of the data and assumptions used to create it. Understanding the data-to-knowledge process includes asking who benefits and who may be disadvantaged.

The central lesson is that organisations create value through a chain rather than through collection alone. Records must be given stable meanings, brought together carefully, interpreted in context, and connected to an accountable decision. Later sections examine the systems that support this chain: governance, databases, analytical environments, knowledge practices, infrastructure, and emerging technologies. Each one addresses a different part of the question: how can data become useful, trusted, and actionable?

02 Managing organisational data



Recommended video: Why dashboards fail without human knowledge

This audio asks why dashboards cannot make sense of metrics on their own, highlighting the importance of context, experience, purpose, and human judgement. [Watch the video](#) ■

READING // ORGANISATIONAL CONTEXT

Managing organisational data begins with a practical problem: the same organisation often holds many versions of the same fact. A customer may appear in a sales system, service platform, finance application, website, and spreadsheet. A product may have several codes or descriptions. An employee's role may be current in one system and outdated in another. These inconsistencies create extra work and make it difficult to know which record should be trusted.

Data quality is not one single property. Data may be inaccurate, incomplete, inconsistent, out of date, duplicated, inaccessible, or poorly defined. A correct address that is five years old is not useful for delivery. A complete customer list may still be misleading if "customer" is defined differently by marketing and finance. Quality must therefore be judged against a purpose, a time period, and the people who need to use the data.

Governance provides the organisational arrangements for dealing with these problems. It identifies ownership, responsibilities, standards, access rules, retention requirements, and processes for resolving issues. Governance asks who is accountable for a definition, who may change a record, how exceptions are handled, and what evidence is needed before data is shared. These are management questions because they affect risk, service, compliance, cost, and trust.

Data stewardship turns broad governance principles into everyday practice. A data steward may document definitions, monitor quality, investigate duplicates, coordinate corrections, and help users

understand the limits of a dataset. Stewardship is not simply checking spelling. It involves negotiating shared meanings between departments that may use the same data for different purposes.

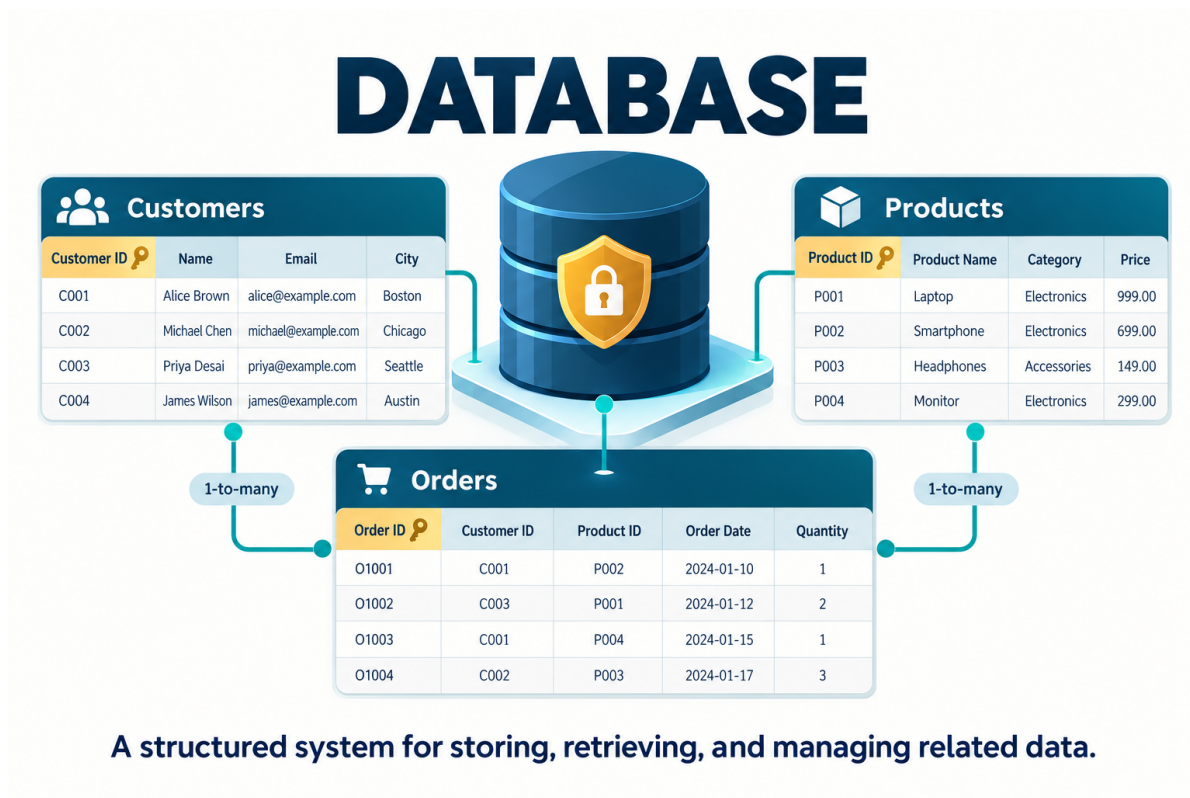
Security and privacy are connected to quality but are not the same thing. A record can be accurate yet exposed to people who should not see it. It can be securely stored yet collected without a clear purpose. Effective management considers confidentiality, integrity, availability, lawful or ethical use, and the rights of the people represented in the data. Controls must be practical enough to work in everyday routines.

Enterprise systems illustrate why management becomes difficult at scale. ERP systems coordinate processes such as purchasing, inventory, finance, production, and human resources. Master Data Management helps keep shared records for customers, products, suppliers, employees, and locations consistent across those processes. ERP and MDM are complementary: one coordinates work, while the other supports trusted shared descriptions of important entities.

Technology cannot solve an ownership problem by itself. A new platform may make it easier to validate an email address, but it cannot decide which department is responsible for the customer record. Automated matching may identify likely duplicates, but people still need to decide whether two records represent the same entity. Artificial intelligence can detect patterns in poor data, but it cannot supply missing accountability or purpose.

Good data management is therefore socio-technical. It aligns systems with policies, workflows, professional judgement, and organisational incentives. Staff need clear procedures and enough time to follow them. Managers need to act on quality issues rather than treating them as someone else's problem. When responsibility is visible and shared, data becomes more reliable throughout its lifecycle instead of deteriorating in disconnected silos.

03 The database approach



Recommended video: What is a Database? — IBM Technology

This video introduces databases as structured systems for storing, relating, retrieving, and protecting records rather than simply collecting more rows. [Watch the video](#) ■

READING // ORGANISATIONAL CONTEXT

A **database** is more than a large electronic filing cabinet. It is a designed representation of an organisation's entities, attributes, **relationships**, and rules. An online shop needs to represent customers, orders, products, payments, and stock. A university needs students, units, enrolments, assessments, and results. The design determines what can be represented clearly and what questions can be answered reliably.

Relational databases organise information into tables and connect those tables with keys. This allows a customer to be stored once while many orders refer to that customer. A product can have one maintained description and price rather than hundreds of copied versions. Reducing unnecessary duplication makes updates safer, but it also requires designers to understand which facts belong to which entity and how entities relate.

Database rules protect the consistency of those relationships. An order may need a valid customer identifier. A payment may need to refer to an existing order. Stock may not be allowed to fall below an agreed limit. These constraints are useful because they prevent obvious errors, but they also formalise organisational assumptions. The database is not neutral: it encodes a view of what counts as a valid record and a valid relationship.

A Database Management System provides the operational tools around the design. It supports queries, transactions, access controls, backups, concurrency, recovery, and administration. Transactions are

especially important when several changes belong together. A sale may need to record the order, reduce stock, authorise payment, and issue a receipt without leaving half-completed records if something fails.

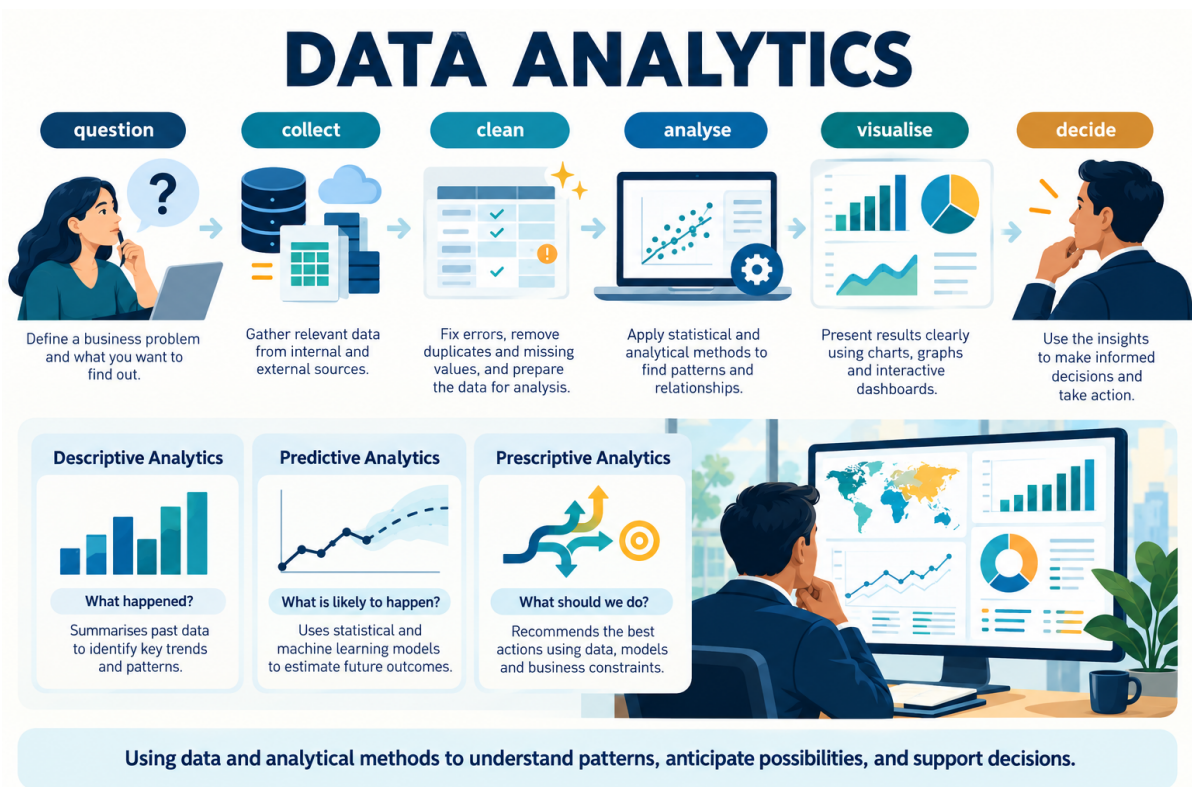
The database approach also changes how people work. Instead of keeping separate personal lists, teams can use shared records and standard queries. This improves coordination, but it can expose disagreements that were previously hidden. Departments may discover that they use different definitions, that important fields are missing, or that a process depends on information nobody formally owns. Database projects therefore reveal organisational issues as well as technical ones.

Not every data problem is best solved with the same database design. Relational systems are strong where relationships, consistency, and transactions matter. Document, key-value, column, and graph approaches may be useful when data structures are flexible, workloads are distributed, or relationships need to be explored differently. The choice should follow the information need rather than a fashion for a particular technology.

Cloud database services make it easier to provision storage, connect applications, expose APIs, and scale capacity. They also create new responsibilities. Organisations still need to design the data model, control access, understand costs, plan backups, manage vendors, and meet privacy obligations. Moving a database to the cloud changes where and how it operates; it does not remove the need for design and governance.

A well-designed database supports trustworthy organisational activity because it makes relationships explicit, enforces important rules, and gives people consistent ways to retrieve information. Its success should be judged by whether it supports the work and decisions around it. More rows, faster hardware, or a newer platform do not compensate for a design that misrepresents the organisation or hides the meaning of its records.

04 Warehouses and decision support



READING // ORGANISATIONAL CONTEXT

Organisations need different information systems for different kinds of questions. A point-of-sale system must record a purchase quickly and accurately. A manager may want to compare sales across stores, months, products, and customer groups. These activities involve related data but place different demands on a system. Confusing them can make **transactions** slow or make analysis too limited to be useful.

Operational systems are designed around current activity. They process orders, payments, bookings, inventory changes, and service requests, often one transaction at a time and with many users working concurrently. Their priorities include speed, availability, consistency, and reliable recovery. They are not usually designed to scan years of activity across many departments for a complex comparison.

Analytical environments are designed around investigation and comparison. A **data warehouse** brings together historical information from operational systems and organises it for reporting and analysis. A data mart focuses on a subject such as finance, marketing, or operations. These environments allow users to ask broader questions without putting the same pressure on the systems that run daily work.

Moving data into a warehouse requires choices. ETL or ELT processes extract data, transform it, and load it into an analytical structure. Dates may need a common format. Product codes may need to be matched. Cancellations, refunds, and missing values need agreed treatment. The resulting dashboard is therefore not a transparent window onto reality; it is the end of a chain of definitions and transformations.

A data lake takes a more flexible approach by storing large amounts of data in varied or near-original forms. It may contain logs, documents, images, sensor streams, click events, and tables. This flexibility supports exploration and machine learning, but it can also create a “data swamp” if nobody knows what

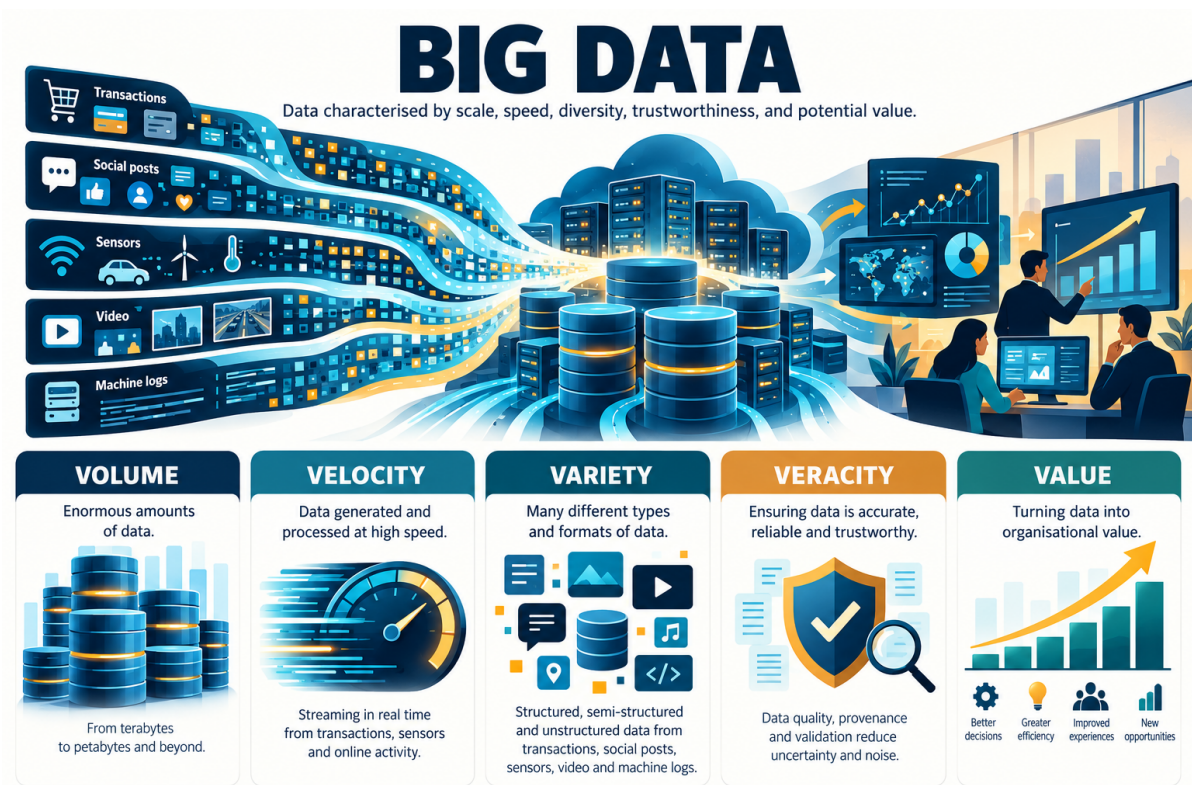
the files mean, where they came from, whether they are current, or who may use them.

Lakehouses attempt to combine the flexibility of a lake with stronger analytical management. Whatever architecture is chosen, metadata and governance remain essential. Users need to understand the source, grain, definitions, time period, transformations, and limitations of a dataset. A polished chart can create false confidence if these details are hidden.

Business intelligence turns prepared data into reports, dashboards, and comparisons. The visual design matters, but interpretation matters more. A line that falls may represent a real change, a reporting delay, a new definition, or a missing source. Analysts need to investigate unusual results and explain the reasoning behind measures rather than presenting every number as self-explanatory.

Decision support is successful when it connects analysis to action. A warehouse may reveal that sales fall after stockouts, that a service is used differently by different groups, or that a process is taking longer over time. People still need to decide what to do, test whether the interpretation is sound, and consider consequences. Analytical systems improve decisions when they preserve context instead of replacing judgement with a dashboard.

05 Big Data and analytics



Recommended video: Big Data — Tim Smith (TED-Ed)

This video uses CERN to explain how enormous datasets changed the way organisations store, share, process, and find value in information. [Watch the video](#) ■

READING // ORGANISATIONAL CONTEXT

Big Data is not simply a fashionable name for a large spreadsheet. It describes information environments in which scale, speed, diversity, uncertainty, and potential usefulness create new organisational challenges. The five Vs—**volume**, **velocity**, **variety**, **veracity**, and **value**—help explain why existing processes may no longer be sufficient. The important question is how the data changes what an organisation must do.

Volume affects storage, movement, processing, and cost. A streaming platform may collect viewing histories, searches, ratings, device records, playback events, and network measurements from a global audience. Even when individual events are small, their accumulation creates a substantial operational and analytical environment. Organisations need ways to retain useful history without assuming that every record deserves permanent storage.

Velocity concerns how quickly data is generated and how quickly a response is needed. A monthly report can tolerate delay, while fraud detection, adaptive streaming, or equipment monitoring may depend on near-real-time processing. The required speed affects pipelines, databases, network design, staffing, and the relationship between operational and analytical systems.

Variety appears when organisations combine structured records with text, images, video metadata, logs, sensor streams, click events, and device information. These sources do not share the same format or meaning. Integration requires common identifiers, metadata, transformation rules, and careful

attention to what can legitimately be compared. More variety can create richer insight, but it can also create more opportunities for misleading combinations.

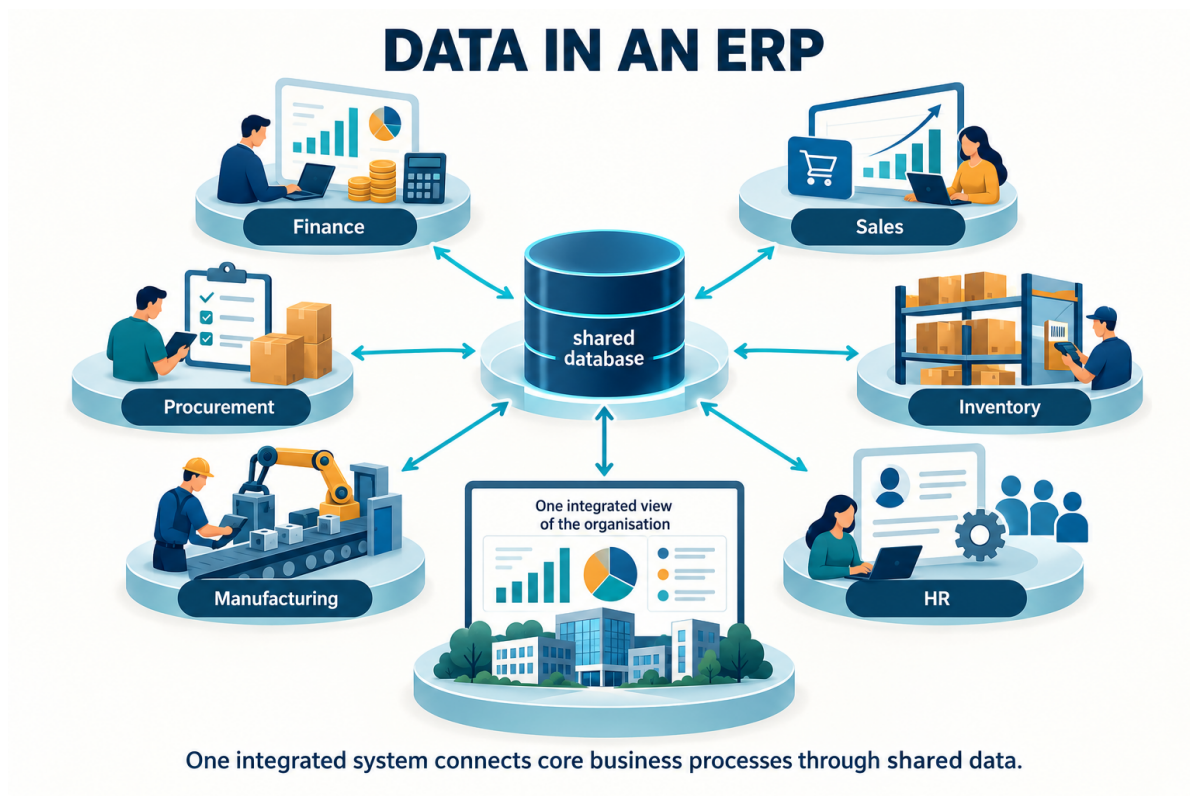
Veracity asks whether the data can be trusted. Records may be missing, duplicated, delayed, biased, affected by shared accounts, or produced by a system that measures only part of the situation. A recommendation based on one person's unusual viewing may not represent a stable preference. Trust requires provenance, validation, monitoring, and an honest description of uncertainty.

Value is the reason organisations invest in Big Data. Value may appear as improved service, better planning, lower costs, new products, safer operations, or more relevant recommendations. It is not guaranteed by scale. Data creates value only when it is connected to a meaningful purpose and when the organisation can act on the result responsibly.

Analytics and artificial intelligence extend this pipeline. Models use selected and prepared data to identify patterns, estimate likely outcomes, rank options, or recommend actions. A model's output is not a fact about the future; it is a prediction shaped by its training data, features, assumptions, and evaluation criteria. Human oversight is needed to decide when an output is useful and when it should be questioned.

Big Data also changes relationships between organisations and the people they observe. Continuous tracking can improve personalisation while making behavioural inferences less visible. Recommendations can help users find relevant content while shaping attention and narrowing exposure. Students should therefore assess both technical performance and questions of privacy, fairness, transparency, autonomy, and who benefits from the data economy.

06 Knowledge management



Recommended video: What is Knowledge Management? — IBM Technology

This video introduces the organisational practices used to capture, share, organise, and apply knowledge across people and systems. [Watch the video](#) ■

READING // ORGANISATIONAL CONTEXT

Knowledge management begins with a simple organisational reality: important know-how is often distributed across people, conversations, routines, examples, and memories rather than stored in one system. An experienced technician may recognise a fault from a sound. A tutor may know which explanation helps students. A project team may remember why a design decision was made. If this knowledge cannot travel, the organisation repeatedly pays to rediscover it.

Knowledge management is not the same as document storage. It involves creating, capturing, organising, finding, sharing, applying, and improving knowledge. A repository can contain thousands of files and still be useless if the files are poorly named, out of date, difficult to search, or disconnected from the work where people need them. Usefulness is a better measure than volume.

The distinction between explicit and **tacit knowledge** is helpful. **Explicit knowledge** can be written, illustrated, indexed, and searched in manuals, procedures, FAQs, case records, and learning materials. Tacit knowledge includes judgement, experience, relationships, and situational understanding. It is difficult to capture completely, so good systems support conversation, mentoring, examples, communities of practice, and opportunities to ask questions.

A knowledge-management system may be a searchable knowledge base, expert directory, lessons-learned repository, collaboration space, case-management platform, wiki, learning system, or recommendation tool. The technology is only part of the system. People need to know why they should

contribute, trust what others have written, and see a clear benefit when they reuse it.

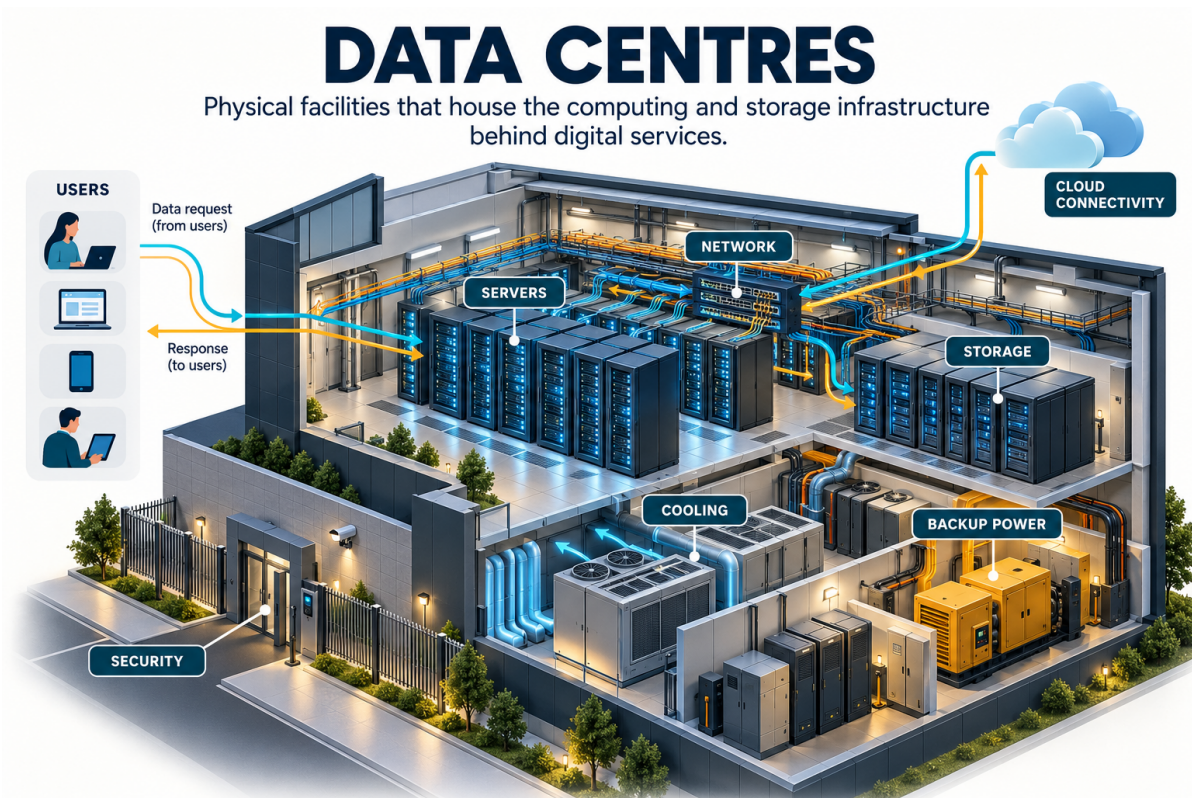
Curation gives knowledge context and durability. A useful article identifies its audience, owner, source, date, scope, and review point. It distinguishes a general rule from a local workaround. It records not only what to do but why, under which conditions, and with what exceptions. Without this context, an organisation may preserve advice that was once correct but is now unsafe or irrelevant.

Knowledge sharing is also shaped by power and culture. Staff may hesitate to document mistakes if they expect blame. Experts may be reluctant to share rare knowledge if recognition and career rewards favour individual ownership. Managers can support learning by making reflection part of work, recognising contributions, and treating questions and corrections as improvements rather than failures.

Search and reuse matter because knowledge that cannot be found is effectively unavailable. Labels, metadata, examples, natural-language search, and links between related cases help users move from a problem to a reliable answer. Even then, users need enough explanation to judge whether an answer applies to their situation. Search ranking should not be mistaken for truth.

Knowledge management succeeds when it helps people do something better: solve a customer problem, onboard a colleague, avoid a repeated mistake, make a safer decision, or adapt a practice to a new situation. It is therefore a social and organisational capability supported by systems. Its success depends on trust, incentives, curation, participation, and continuous learning—not on the size of the repository.

07 Data centres and the cloud



Recommended video: Scientists Are Creating The World's First Biological Computer In Australia

This video introduces biological computing with living neurons and connects emerging hardware to the energy demands of artificial intelligence. [Watch the video](#) ■

READING // ORGANISATIONAL CONTEXT

Cloud services can feel weightless because users see an interface rather than the infrastructure behind it. When someone streams a film, submits an assignment, opens a stored document, or uses an AI service, the request travels through networks to computing and storage resources. The cloud does not remove physical infrastructure; it makes the physical layer less visible to the person using the service.

A **data centre** combines many systems that must work together. Servers execute workloads. Storage systems hold files and databases. Networks move requests and responses. Power distribution keeps equipment running. Cooling removes heat. Monitoring, security, backup, and skilled operations help maintain availability. A failure in any of these layers can affect the digital service that users experience.

Cloud providers allow organisations to rent capacity rather than owning every server. This can provide elasticity: an organisation may increase capacity during a peak and reduce it later. It can also provide access to specialist services, global networks, managed databases, and security tooling. The convenience changes the organisation's responsibilities rather than eliminating them.

Cloud decisions involve contracts, service levels, data location, access controls, recovery arrangements, vendor dependence, and cost. A provider may offer impressive technical capacity, but an organisation still needs to decide which workloads belong there, how sensitive information should be handled, and what happens if the provider suffers an outage or changes its prices or policies.

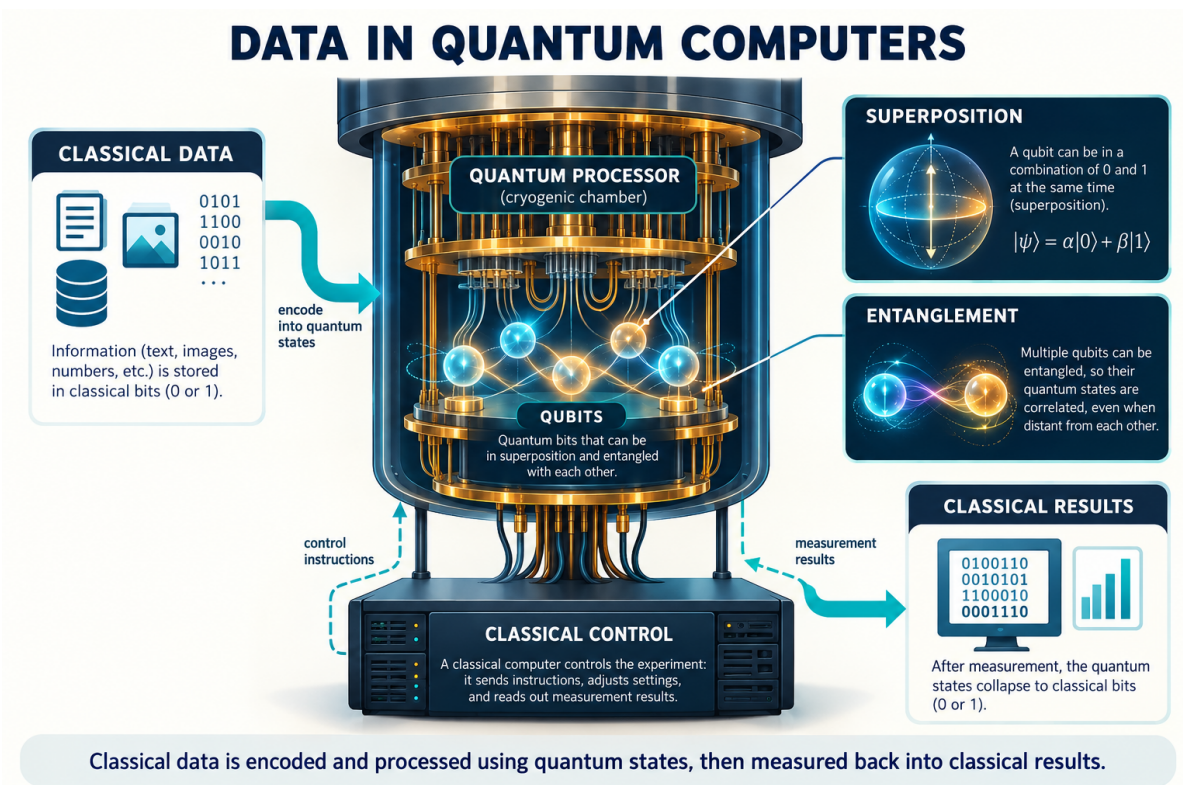
Physical infrastructure has material consequences. Data centres require electricity, land, network connections, construction materials, equipment supply chains, cooling, security, and people. Some cooling systems use water; others use air or specialised technologies. Efficiency improvements can reduce resource use per computation, but total demand may still grow as more services and AI workloads are added.

Location affects both engineering and public policy. A site may offer renewable electricity, fibre connectivity, available land, or a favourable climate. It may also place pressure on a local grid, water supply, road network, or community. Decisions about incentives, approvals, employment, and environmental impact involve people beyond the technology department.

Resilience means more than having a backup server. Organisations need redundant power, networks, storage, facilities, people, procedures, and suppliers. They need to know how long a service can be unavailable, how much data can be lost, and which functions must be restored first. These choices determine the real meaning of “available” in a cloud service.

The cloud therefore raises questions of visibility and responsibility. Users may experience a fast and convenient service without knowing where data is stored, how much energy is used, or which providers are involved. Good information-systems analysis connects the user experience to the infrastructure, contracts, communities, and environmental trade-offs that make the experience possible.

08 Quantum computing: the next frontier



Recommended video: Remote quantum entanglement of nuclear qubits — UNSW

This video explains a significant quantum-computing advance: entanglement between distant nuclear qubits in silicon, a key resource for future quantum systems. [Watch the video](#) ■

READING // ORGANISATIONAL CONTEXT

Quantum computing is best understood as a specialised extension of information processing, not as a replacement for every ordinary computer. Classical computers use bits and well-established architectures. Quantum computers use **qubits** whose states can be prepared and manipulated according to quantum-mechanical rules. The practical question is not whether quantum sounds powerful, but which problems may benefit from its particular computational resources.

Superposition allows a qubit to be described using a combination of possible states before measurement. **Entanglement** creates correlations between qubits that cannot be represented as independent classical states. These ideas do not mean that a quantum computer simply tries every answer and reveals the best one. Algorithms must use interference and measurement so that useful outcomes become more likely.

The technology is difficult because qubits are fragile. Heat, vibration, electromagnetic effects, imperfect control, and interactions with the environment can destroy the state needed for a calculation. Researchers need specialised hardware, calibration, isolation, measurement, error mitigation, and eventually error correction. The supporting classical computers remain essential for controlling the experiment and processing results.

Error correction changes the scale of the challenge. A useful logical qubit may require many physical qubits and substantial monitoring. The number depends on the hardware quality, algorithm, and error

rates. This means that a promising laboratory demonstration does not automatically indicate a dependable business service. Students should separate a physical result, a prototype algorithm, a useful benchmark, and an operational capability.

Possible organisational applications include simulation of materials and chemicals, selected optimisation problems, scheduling, logistics, portfolio analysis, and cryptography. These are possibilities rather than universal benefits. A quantum method must outperform available classical methods for a problem that matters, under realistic data, cost, latency, and reliability constraints. Hybrid systems are more likely than isolated quantum machines to shape early adoption.

Quantum computing also affects information security. Some future quantum algorithms could threaten widely used public-key cryptography, even if large fault-tolerant machines are not yet available. Organisations need to understand where cryptography is used, how long protected information must remain confidential, and whether migration to quantum-resistant approaches will take time. This is a governance and planning issue as well as a physics issue.

Organisations should therefore ask disciplined questions before investing. What problem needs solving? Is it genuinely difficult for current methods? What evidence shows a quantum approach could help? What data and skills are required? How would the result integrate with existing systems? What are the costs, security implications, and opportunity costs of experimentation? These questions protect decision-makers from both hype and premature dismissal.

Quantum computing illustrates a wider lesson about emerging technology. A new capability becomes organisationally useful only when it can be connected to reliable data, suitable algorithms, infrastructure, skills, governance, and a meaningful decision. The future may bring important advances, but responsible analysis distinguishes demonstrated capability from aspiration and asks what evidence would justify moving from research to practice.